BUSINESS INTELLIGENCE (SECP3213)

ALTERYX LAB EXERCISE 1: COMPLETE ETL WORKFLOW

CHAPTER 4 (PART 2)

SEM 2, 2024/2025

STUDENT GUIDELINE

Objective: Build a complete ETL workflow in Alteryx Designer using provided datasets

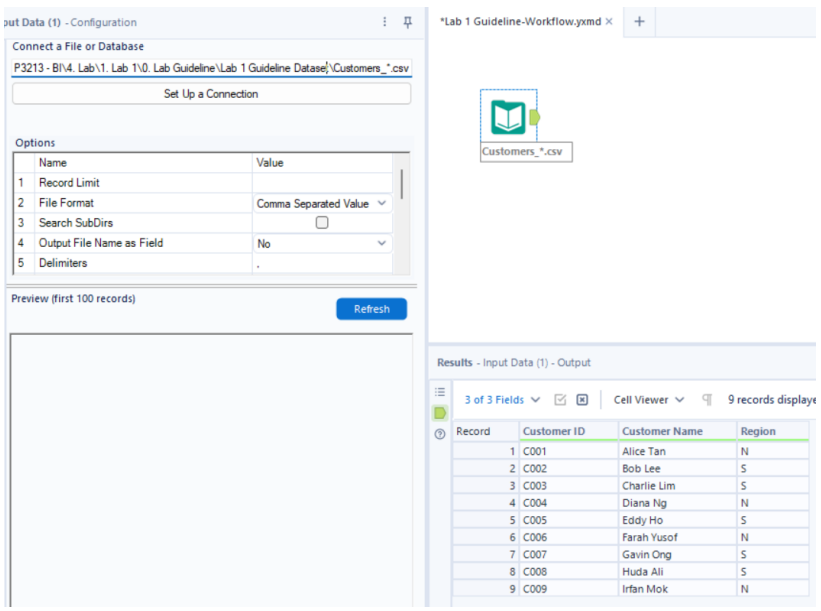
Step 1: Import Data

Tools: Input Data, Dynamic Input, Browse

Reading multiple files using wildcard (*.csv)

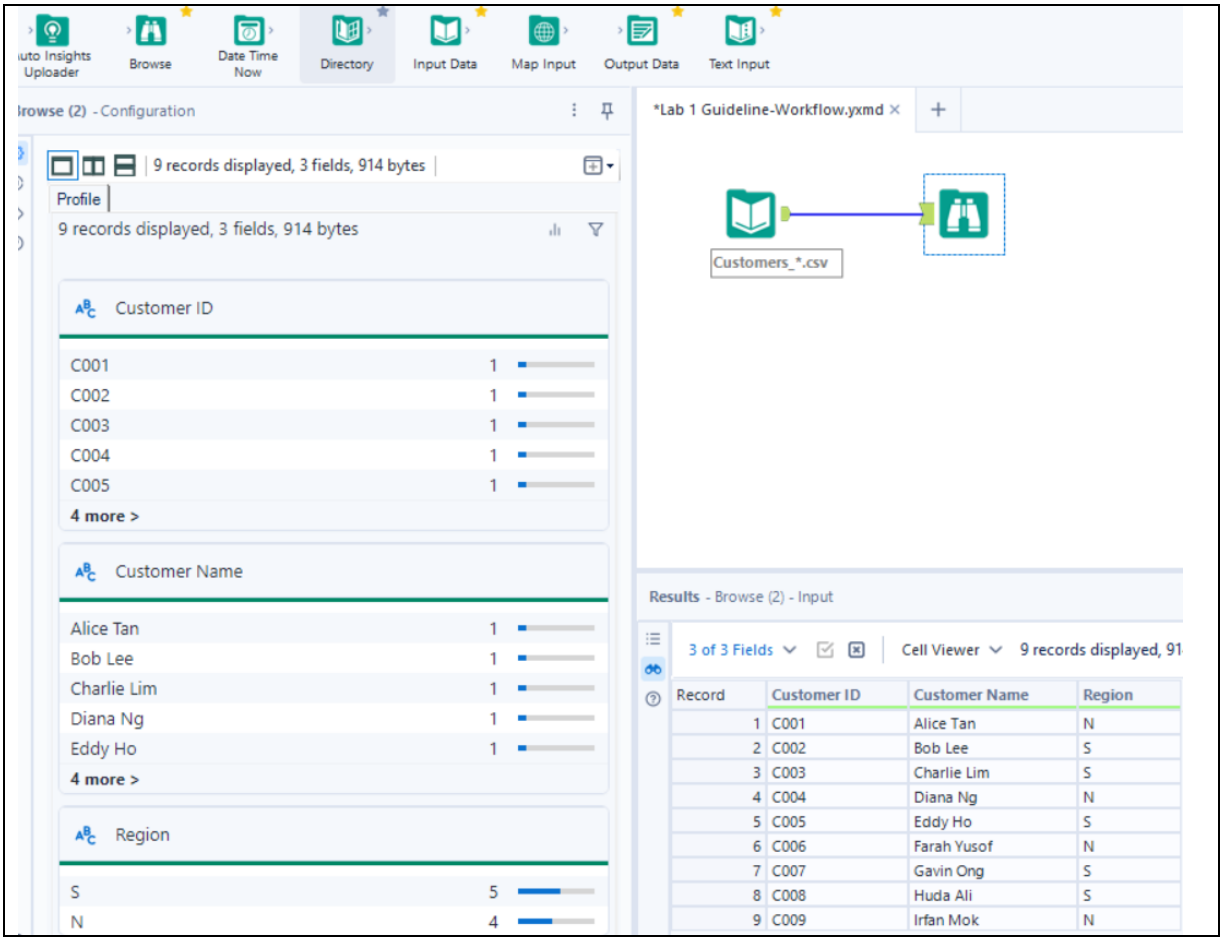
a) Use **Input Data Tool** to import three separate customer CSV files.
*Purpose tool: To **demonstrate reading multiple files** using the Input Data Tool with a wildcard (*.csv).*

- In the file path, select one file, then edit it to:
path_to_folder/Customers_*.csv



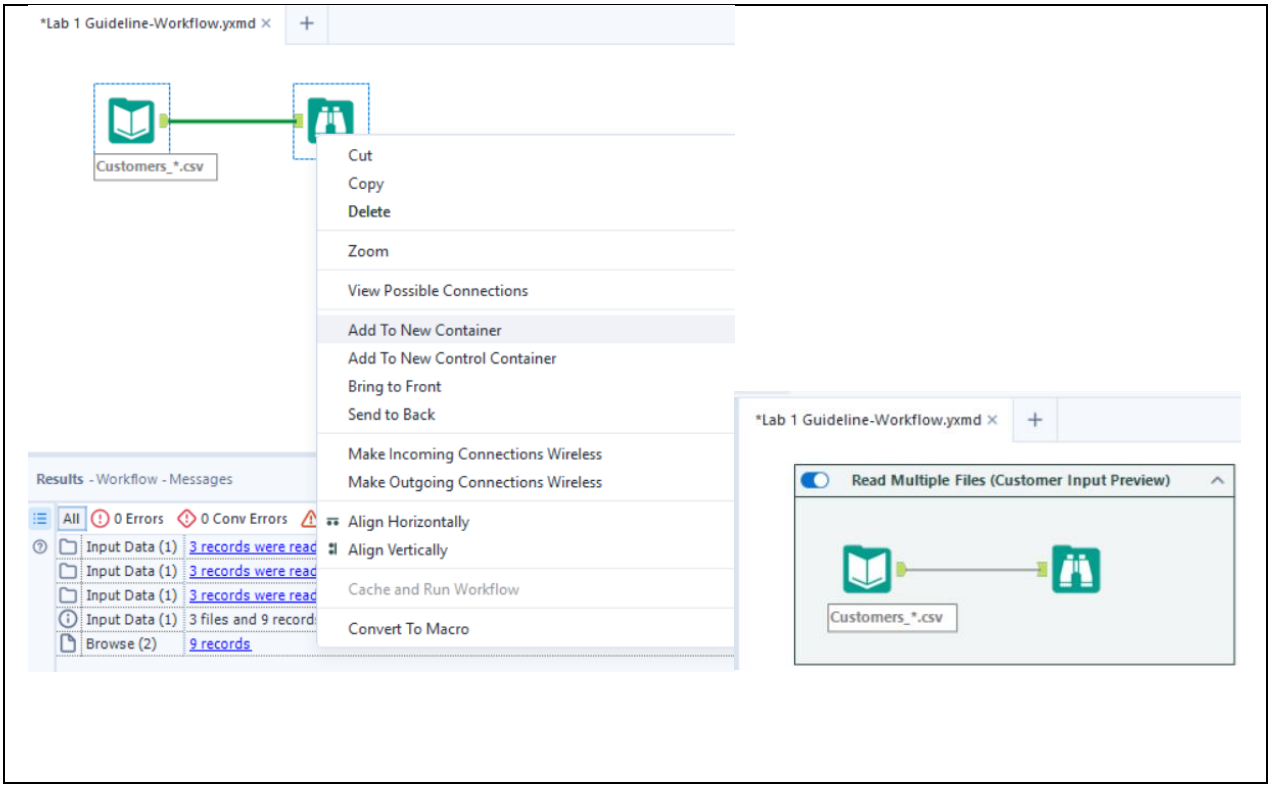
b) Use **Browse Tool** after each input to preview the data.
Purpose tool: to view data from a connected tool (can view profile data in details)

- Details of the data is displayed in the profile tab.



c) Use a **Tool Container** to group the **Input Data Tool** and **Browse Tool** together
Purpose tool: To enable or disable this input section without affecting the rest of the workflow.

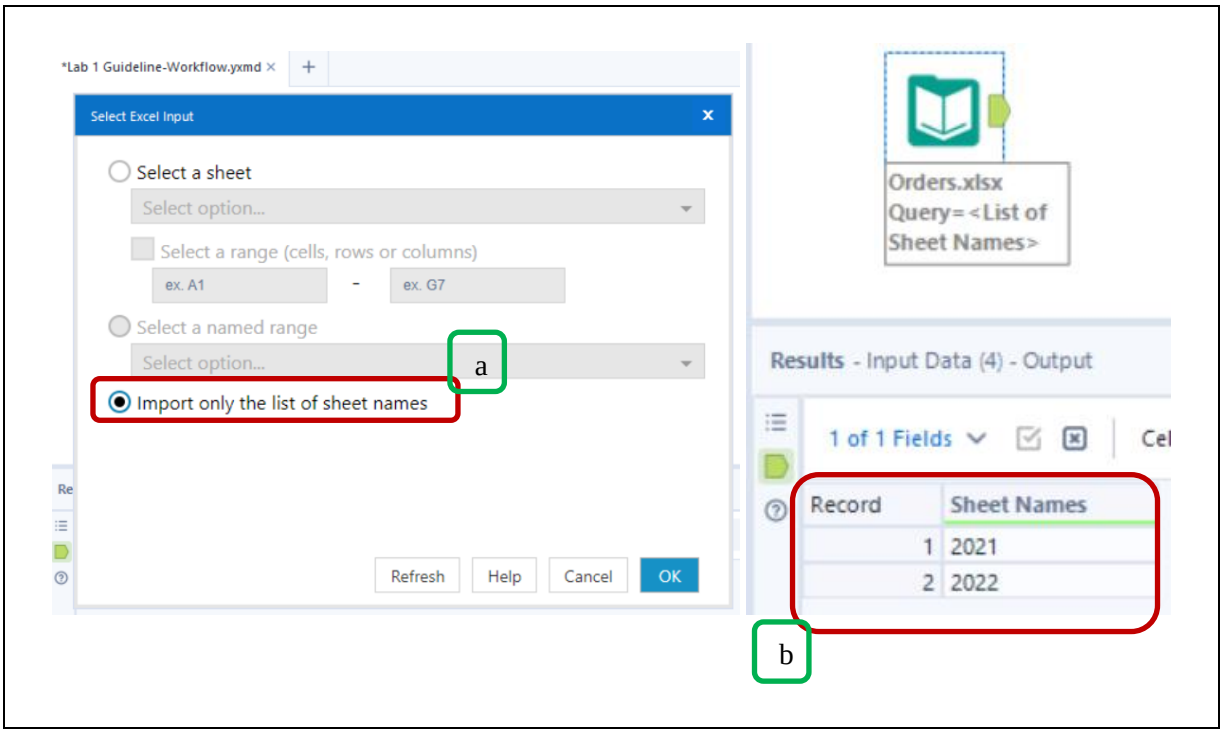
- Highlight the selected tool > right click (*Add to New Container)*
- Rename the container (e.g., “Read Multiple Files-Customer Input Preview”) by double-clicking the label.
- Toggle the checkbox at the top-left corner to enable/disable that group.



Read multiple sheets in a file

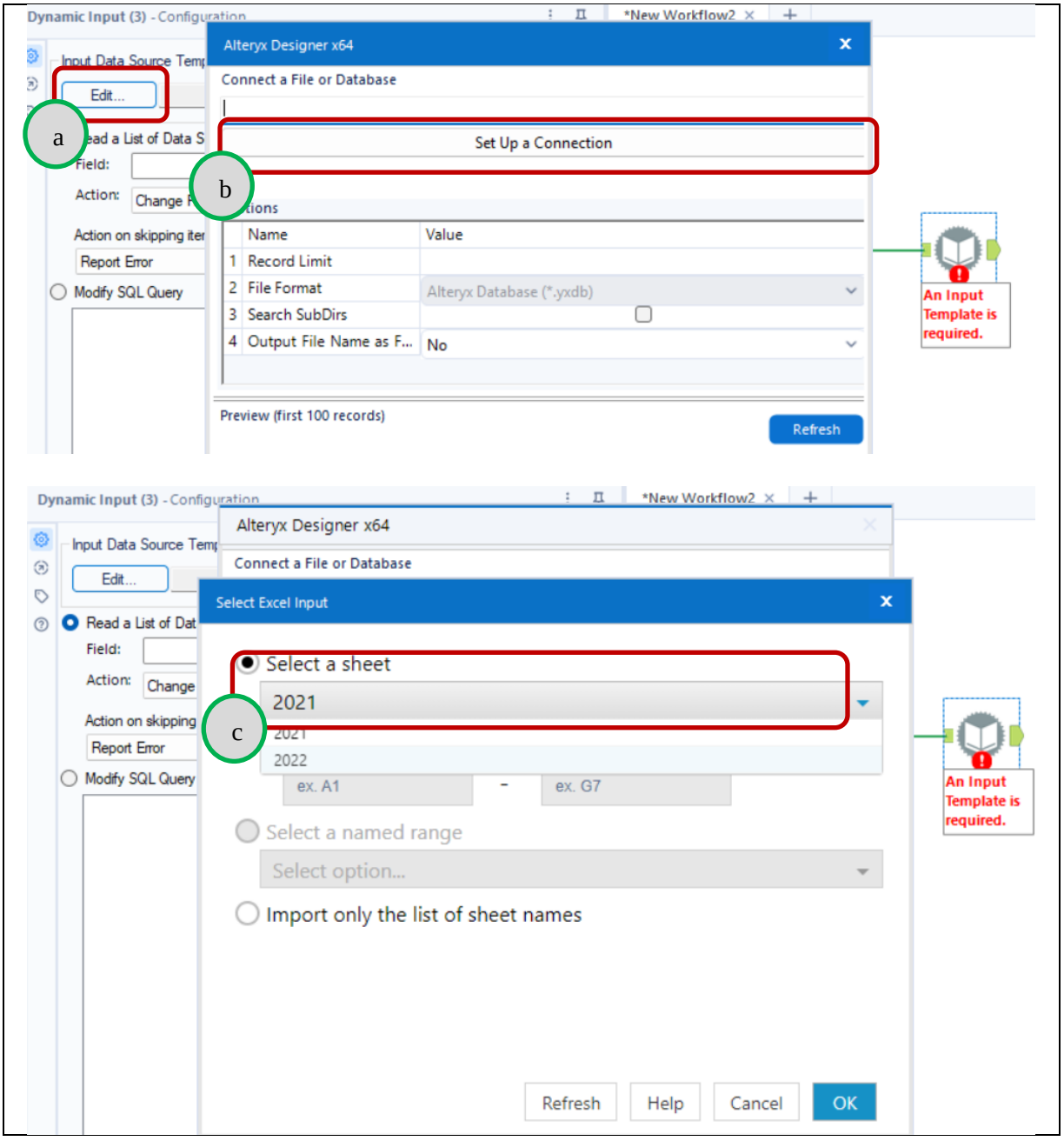
d) Use **Input Data Tool** to import all sheets in Orders.xlsx (2021 and 2022 excel tab).

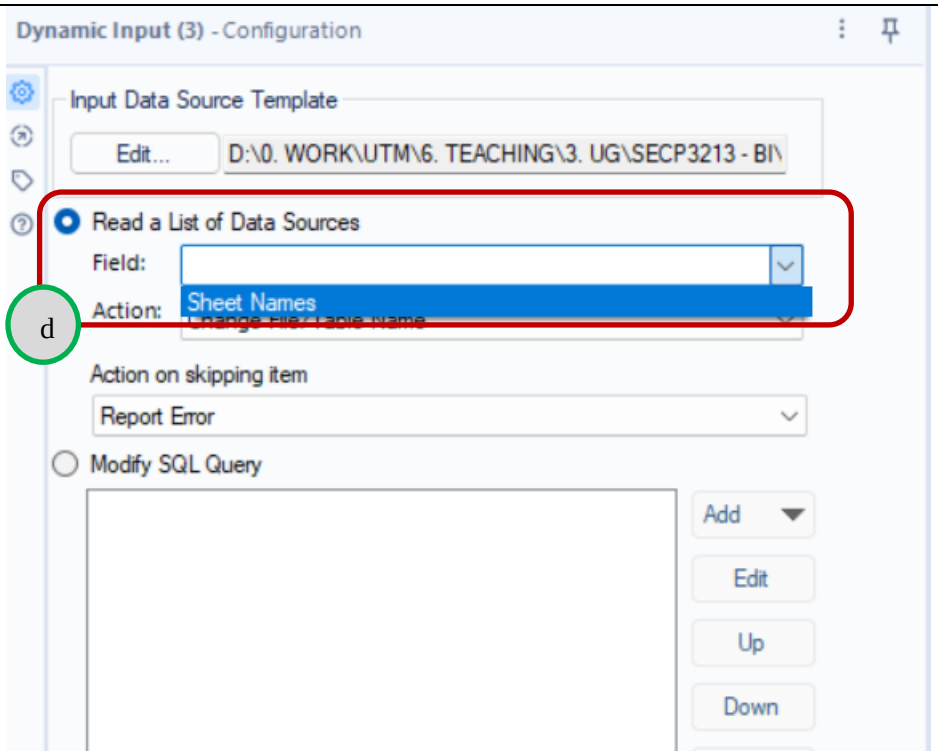
- Configuration of Input Data tool
- Using Order.xlsx file to import data

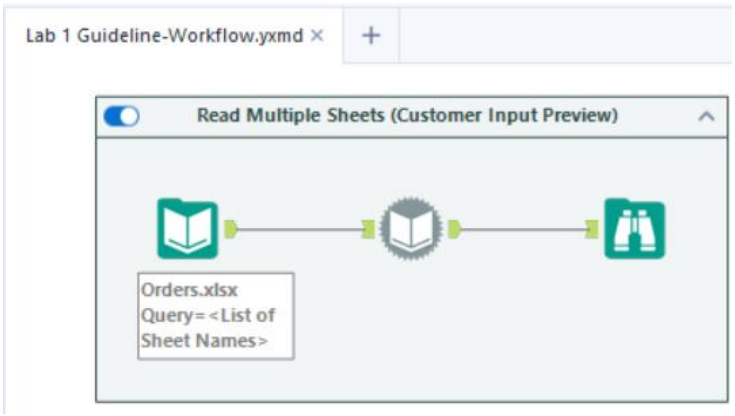


e) Use **Dynamic Input Tool** to read all sheets in Orders.xlsx (2021 and 2022).
*Purpose tool: To read **multiple sheets in 1 file**.*

- Follow step a – b.
- Rename the container (e.g., "Read Multiple Sheets-Customer Input Preview") by double-clicking the label.
- Toggle the checkbox at the top-left corner to enable/disable that group




Dynamic Input (3) - Configuration
Input Data Source Template
Edit...
D:\0. WORK\UTM\6. TEACHING\3. UG\SECP3213 - BI\
Read a List of Data Sources
Field:
Sheet Names
Action:
d
Action on skipping item
Report Error
Modify SQL Query
Add
Edit
Up
Down


Lab 1 Guideline-Workflow.yxmd
Read Multiple Sheets (Customer Input Preview)
Orders.xlsx
Query=<List of
Sheet Names>

Step 2: Data Preparation

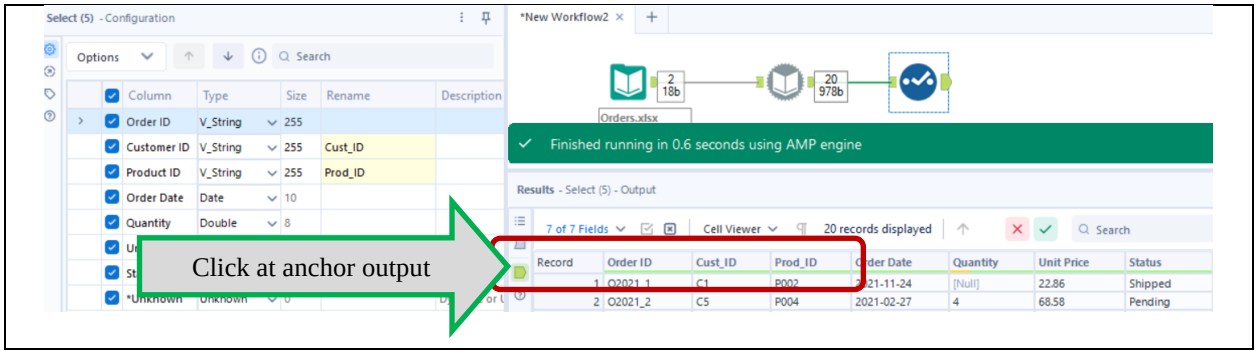
Purpose: Remove mistakes and prepare accurate data

Tools: Select, Imputation, Unique, Filter, Formula (Multi-field, Multi-Row), Sort, Sample, Auto-field

a) Use **Select Tool** to rename columns and fix data types.

Purpose tool: to edit data type, rename field, change size, adding description, and rearrange / sort the order of the data.

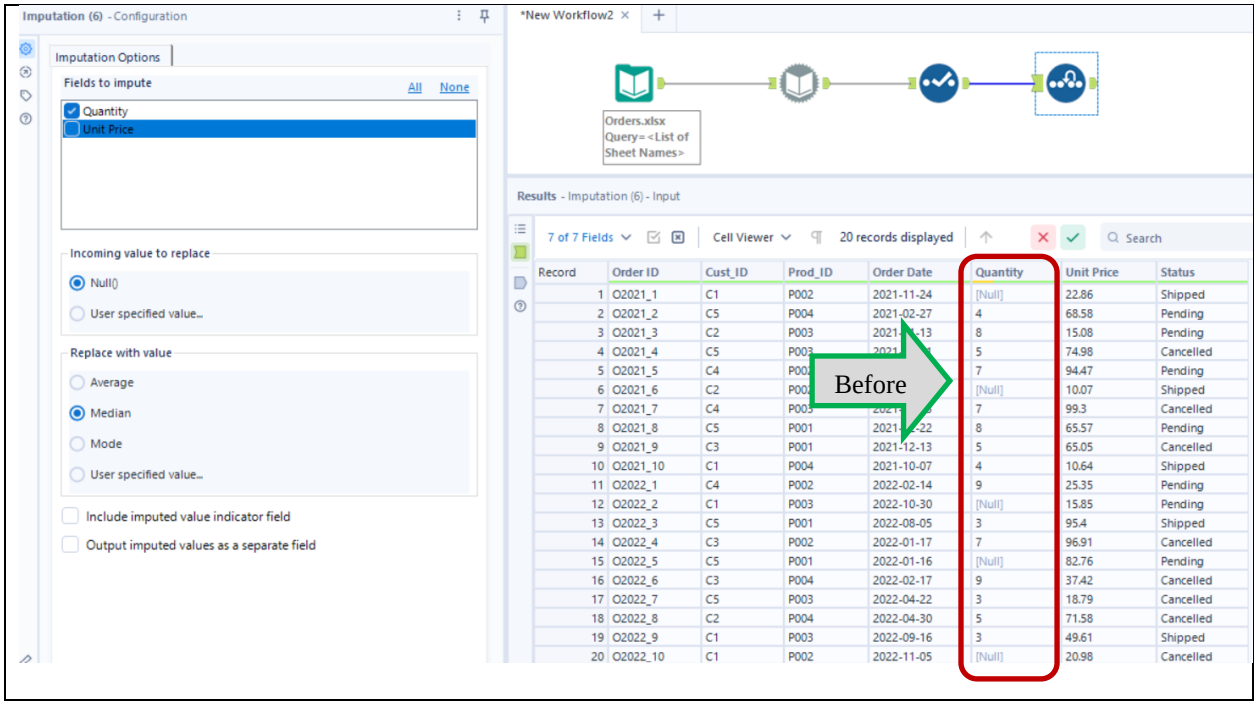
- Using Order.xlsx dataset.
- Rename fields ("Product ID" → "Prod_ID")
- Rename fields ("Customer ID" → "Cust_ID")

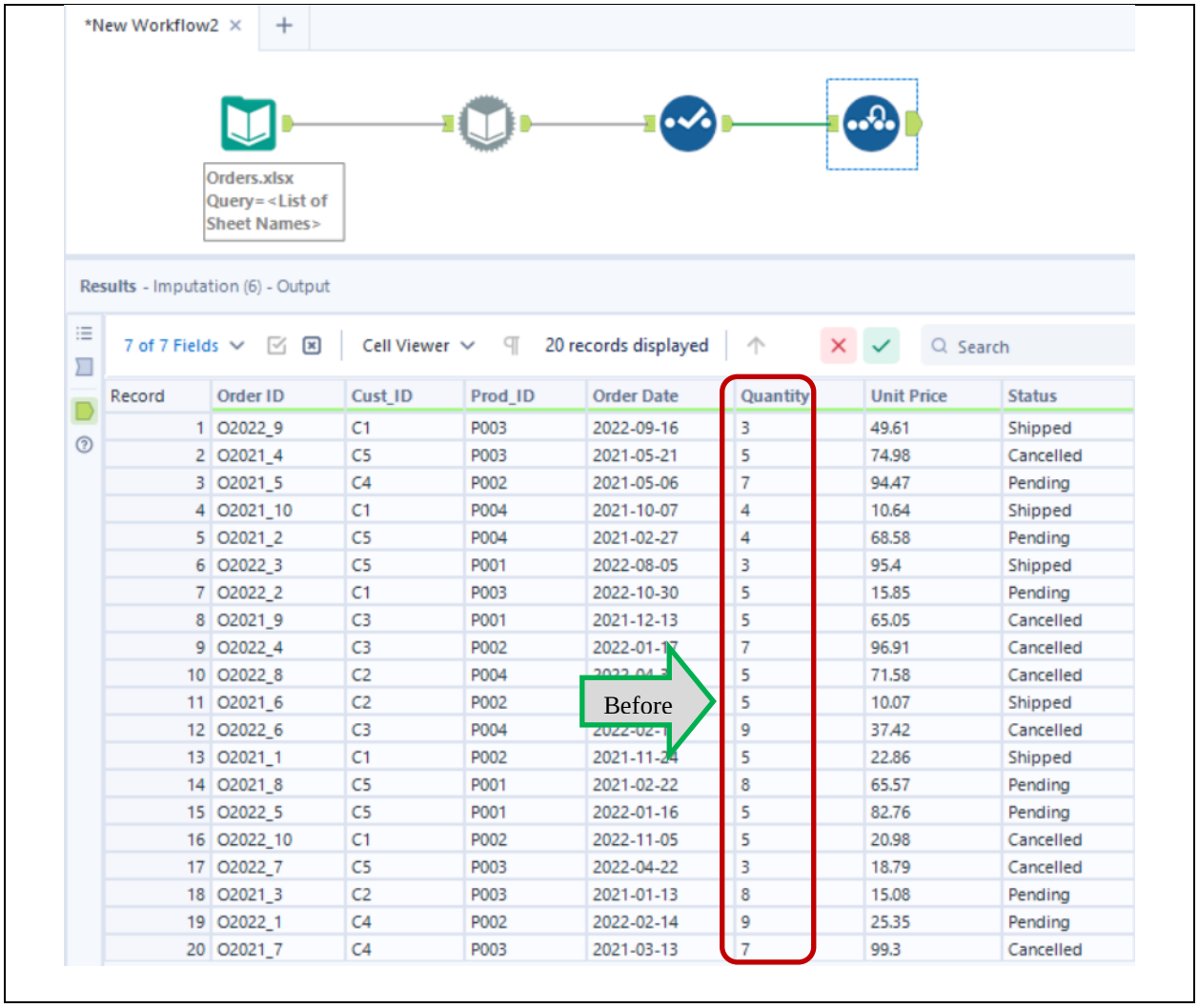


b) Use **Imputation Tool** to fill missing **Quantity** values with the median.

Purpose tool: to replace a specified value within one or more numeric data fields with another specified value (fill missing values).

- Using Order.xlsx dataset.
- Select Quantity field and replace NULLS with Median



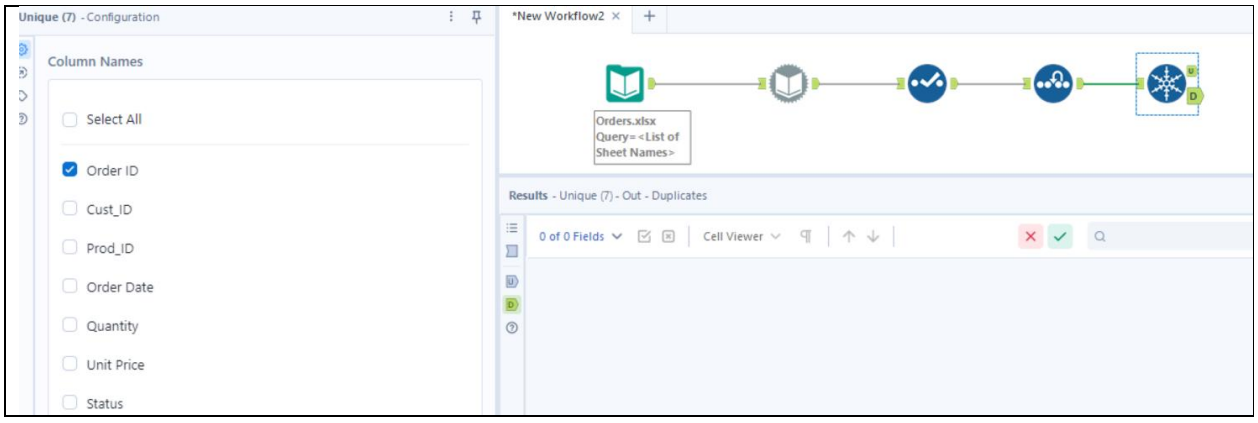


Record	Order ID	Cust_ID	Prod_ID	Order Date	Quantity	Unit Price	Status
1	O2022_9	C1	P003	2022-09-16	3	49.61	Shipped
2	O2021_4	C5	P003	2021-05-21	5	74.98	Cancelled
3	O2021_5	C4	P002	2021-05-06	7	94.47	Pending
4	O2021_10	C1	P004	2021-10-07	4	10.64	Shipped
5	O2021_2	C5	P004	2021-02-27	4	68.58	Pending
6	O2022_3	C5	P001	2022-08-05	3	95.4	Shipped
7	O2022_2	C1	P003	2022-10-30	5	15.85	Pending
8	O2021_9	C3	P001	2021-12-13	5	65.05	Cancelled
9	O2022_4	C3	P002	2022-01-17	7	96.91	Cancelled
10	O2022_8	C2	P004	[illegible]	5	71.58	Cancelled
11	O2021_6	C2	P002	[illegible]	5	10.07	Shipped
12	O2022_6	C3	P004	[illegible]	9	37.42	Cancelled
13	O2021_1	C1	P002	2021-11-24	5	22.86	Shipped
14	O2021_8	C5	P001	2021-02-22	8	65.57	Pending
15	O2022_5	C5	P001	2022-01-16	5	82.76	Pending
16	O2022_10	C1	P002	2022-11-05	5	20.98	Cancelled
17	O2022_7	C5	P003	2022-04-22	3	18.79	Cancelled
18	O2021_3	C2	P003	2021-01-13	8	15.08	Pending
19	O2022_1	C4	P002	2022-02-14	9	25.35	Pending
20	O2021_7	C4	P003	2021-03-13	7	99.3	Cancelled

c) Use **Unique Tool to remove duplicate Order IDs**.

Purpose tool: to distinguish whether data record is unique or a duplicate by grouping on one or more specified fields, then sorting.

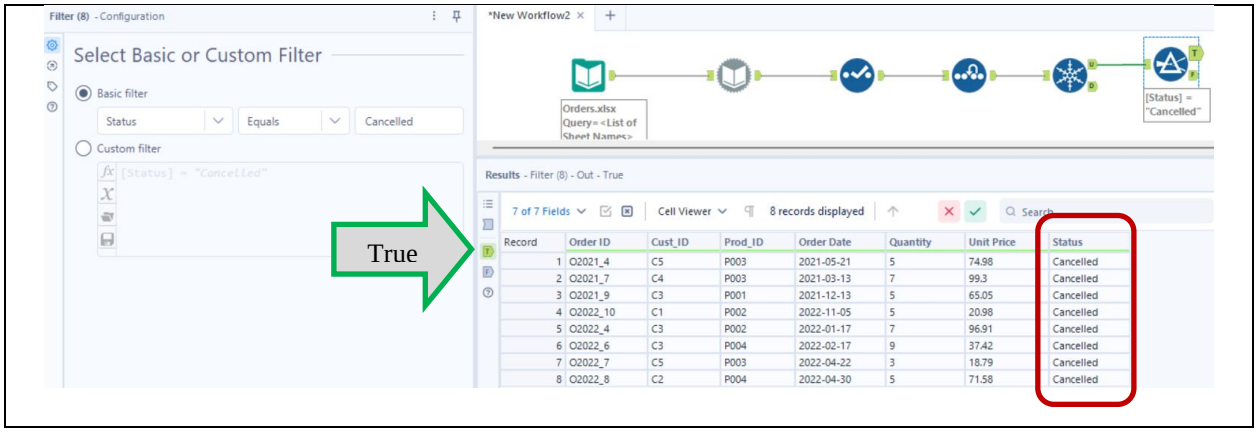
- Using Order.xlsx dataset.
- Remove duplicate Order IDs *(if any)*



d) Use **Filter Tool keep only rows where Status = "Cancelled"**.

*Purpose tool: select data using a condition. Rows of the data that meet the condition ae output to the **T**rue anchor while rows of data that do not meet the condition are output to the **F**alse anchor.*

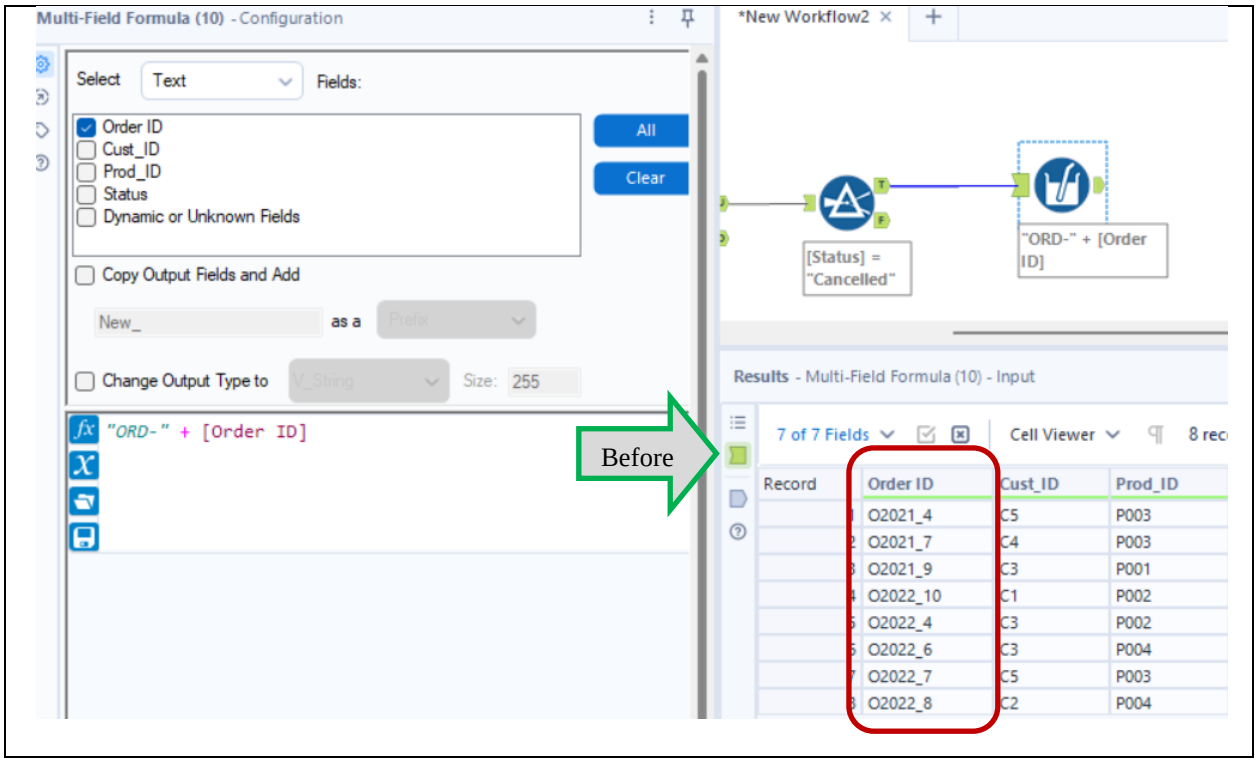
- Using Order.xlsx dataset.
- Keep rows that meet the condition→ Cancelled.
- Choose basic filter (Status → Cancelled), appear in True anchor

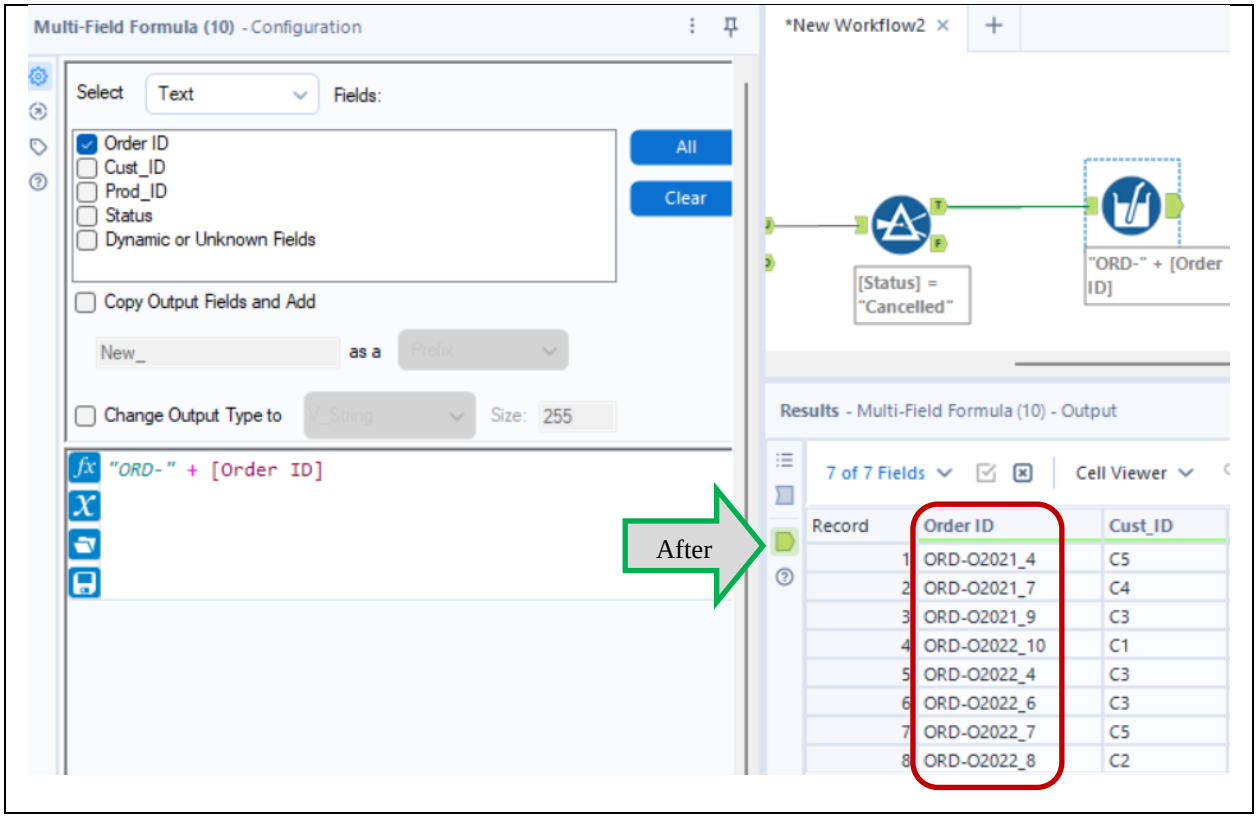


e) Use **Formula Tool (Multi-field) to Add a prefix to IDs.**

Purpose tool: to create or update multiple fields using a single expression.

- Using Order.xlsx dataset.
- Apply to Order ID.
- Expression:
  "ORD-" + [CurrentField_]

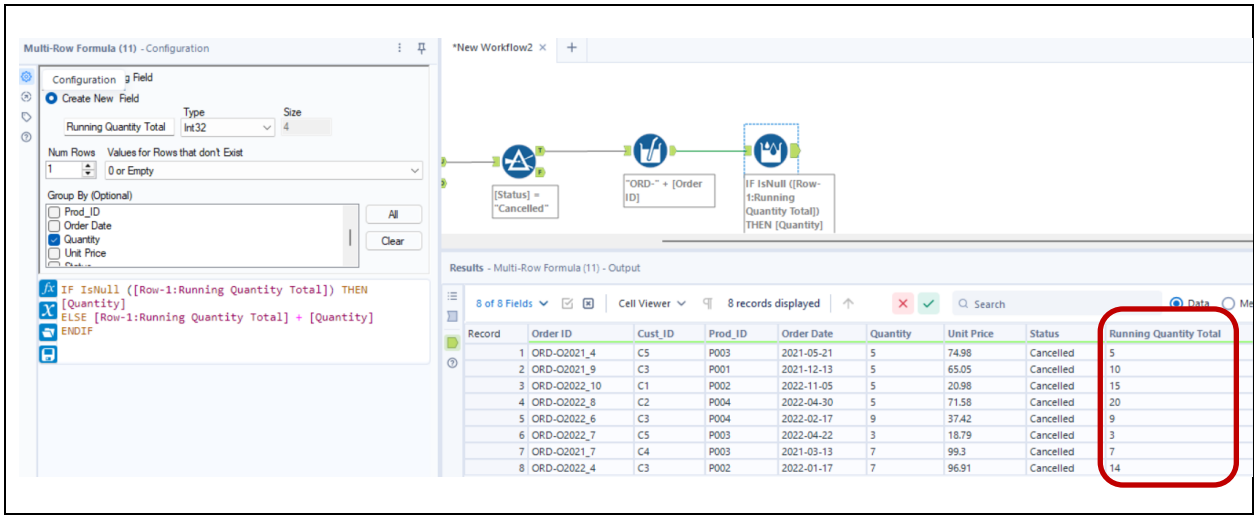




f) Use **Formula Tool (Multi-row) to Calculate Running Total for Quantity.**

Purpose tool: to take the concept of the formula tool a step further (running totals, averages, percentage and other mathematical calculation).

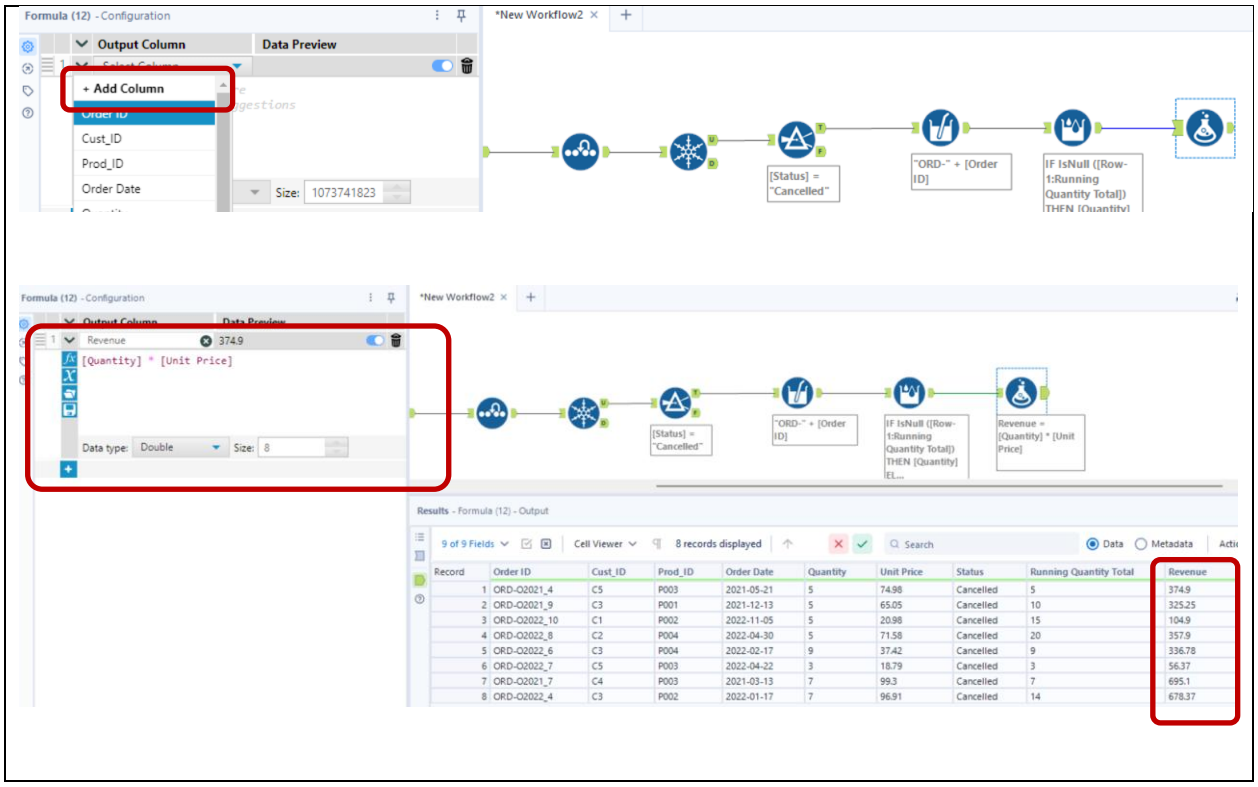
- Using Order.xlsx dataset.
- Create new field: **Running Quantity Total**.
- Group by: **Quantity**
- Expression:
  IF IsNull([Row-1:Running Quantity Total]) THEN [Quantity]
  ELSE [Row-1:Running Quantity Total] + [Quantity]
  ENDIF



g) Use **Formula Tool to create a new calculated field called Revenue using Quantity * Unit Price.**

Purpose tool: to create new column, update column and use one or more expression to perform a variety calculations and operations.

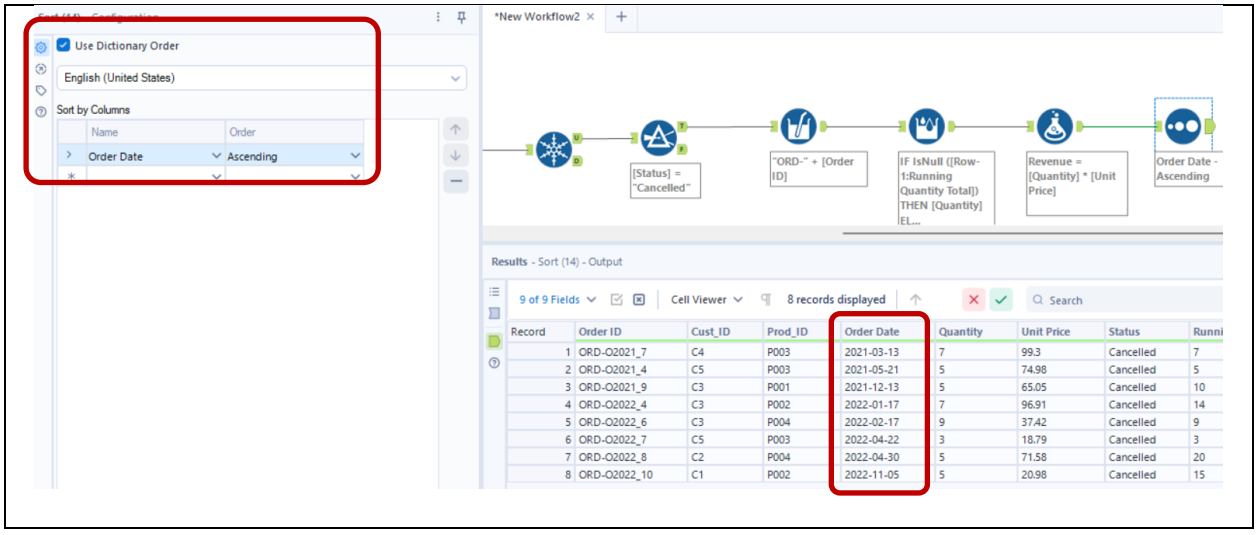
- Using Order.xlsx dataset.
- Drag a Formula Tool after the Multi-Row Formula output.
- Create a new field called **Revenue**.
- Expression:
  [Quantity] * [Unit Price]
- Output type: **Double** (decimal number)



h) Use **Sort Tool to sort data by Order Date**

Purpose tool: to arrange the records in a table in alphanumeric order, based on the values of the specified data fields (arrange rows by name/date).

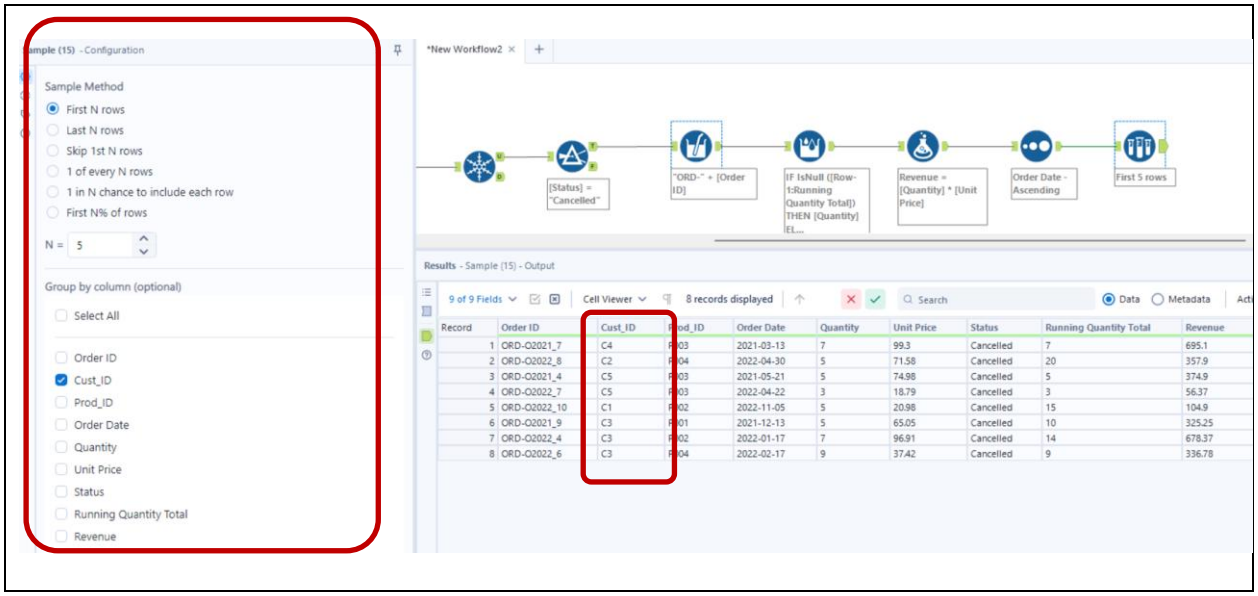
- Using Order.xlsx dataset.
- Sort by Order Date, ascending.



i) Use **Sample Tool to Create a working mini-dataset**

Purpose tool: to limit the data stream to a specified number, percentage or random set of rows. This tool applies the selected configuration to the columns selected to group by.

- Using Order.xlsx dataset.
- Take First 5 records.
- Group by Cust_ID



j) Use **Auto-Field Tool to Optimize field data types.**

Purpose tool: to read through all the records of an input and set the field type the smallest possible size relative to the data contained within the column.

- Using Order.xlsx dataset.
- Automatically adjust fields after Sample.
  - ➢ Numeric fields (e.g., Quantity, Revenue) optimized (e.g., from Double ➔ Int16).
  - ➢ Text fields optimized without excess length.

Step 3: Parse

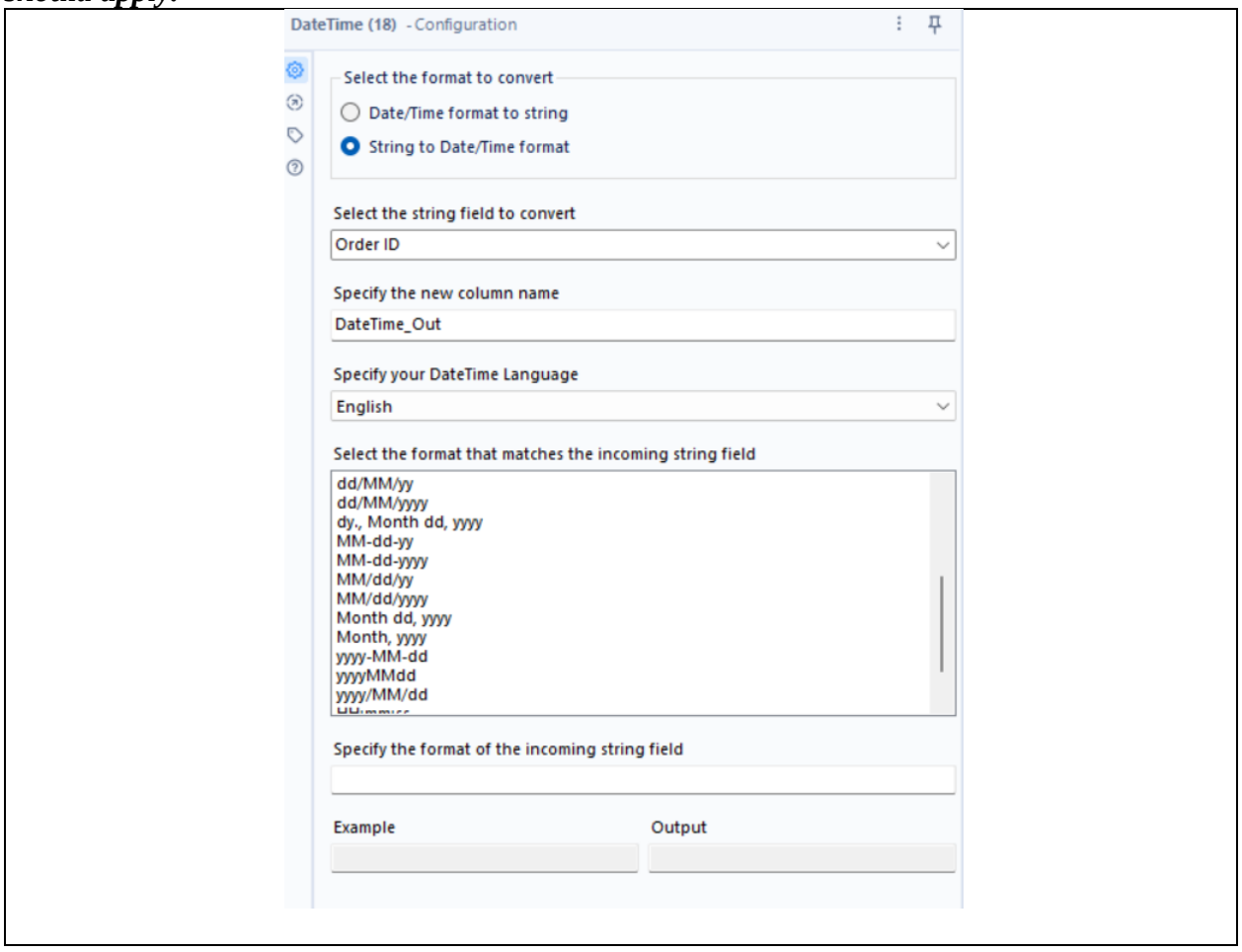
Purpose: Breaking complex or messy data into smaller, meaningful parts, OR Converting data from one format into a cleaner, usable format.
Tools: DateTime, Text to Column

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k) Use **DateTime Tool to Convert Order Date from string to proper Date/Time format.**
Purpose tool: to transform date-time data to and from variety of formats, including both exoression-friendly and human readable formats.

- Using Order.xlsx dataset.
- Drag a DateTime Tool after the cleaning section (after Auto Field or Sample).
- Configure it:
  - Field to convert: Order Date
  - Action: Convert String to Date/Time
  - Format: yyyy-MM-dd (or match your data)
  - New field name: (Optional) Parsed Order Date

Since the date already in datetime format so here, I just attach the overview of details you should apply.

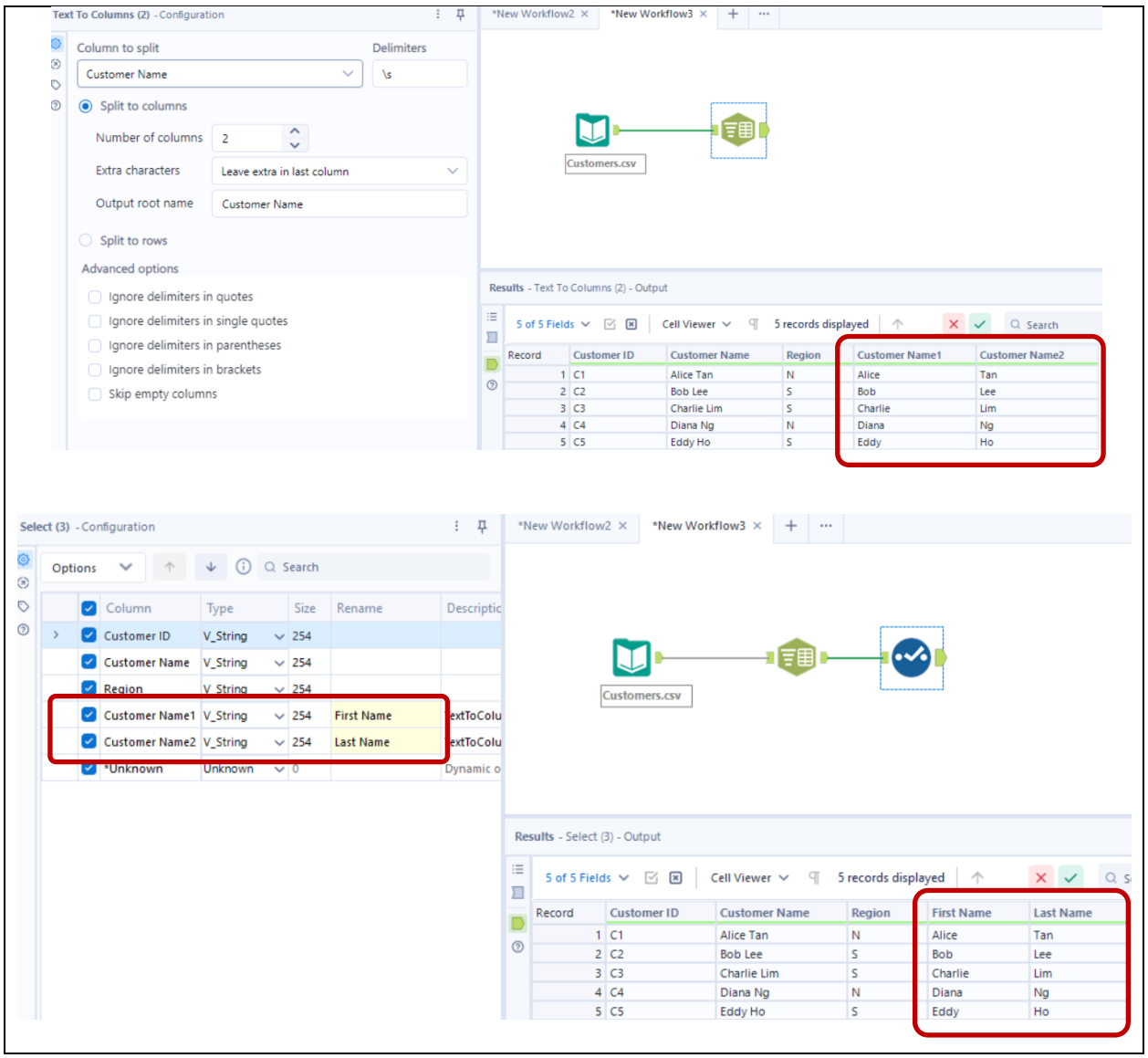


l) Use **Text to Columns Tool to Split Customer Name into First Name and Last Name.**
Purpose tool: to take the text in one column and split the string value into separate, multiple columns (or rows) based on a single or multiple delimeters.

- Using Customer.csv dataset.
- Drag a Text to Columns Tool after the parsed dates.
- Configure it:

- Field to split: **Customer Name**
- Delimiter: Space (one space character)→ \s
- Number of columns: 2
- Direction: Rows or Columns (default is fine)

- *After splitting, the columns created will be named like:*
  - *Customer Name_1*
  - *Customer Name_2*
- Use **Select Tool** to rename the column to **First Name and Last Name**



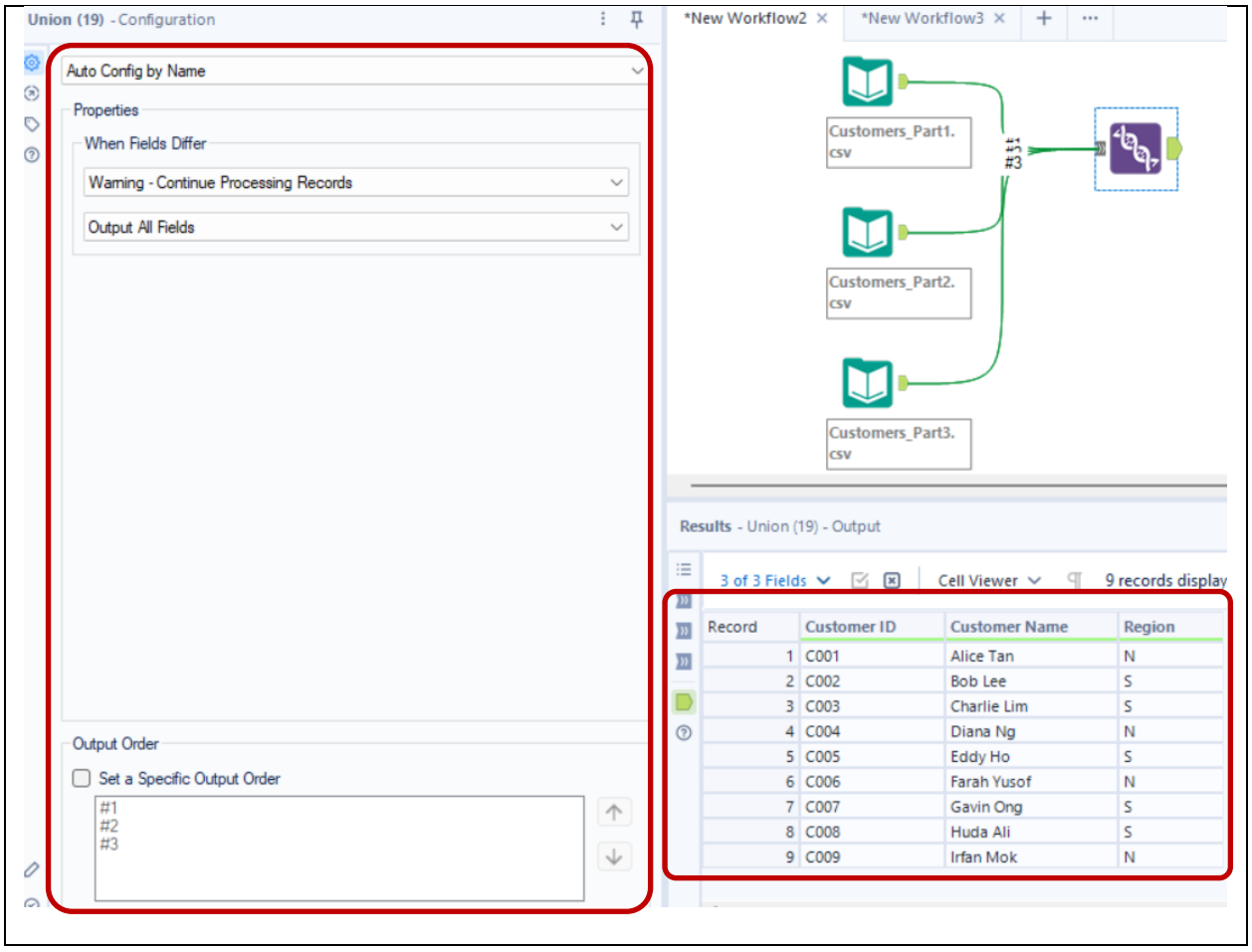
Step 4: Combine Data

Purpose: Join multiple tables or files together into one useful dataset.
Tools: Join, Union, Find and Replace, Append

m) Use **Union Tool** to Combine multiple customer datasets (Customers_Part1, Customers_Part2, Customers_Part3) into one list.

Purpose tool: to combine 2 or more dataset on columns names or position.

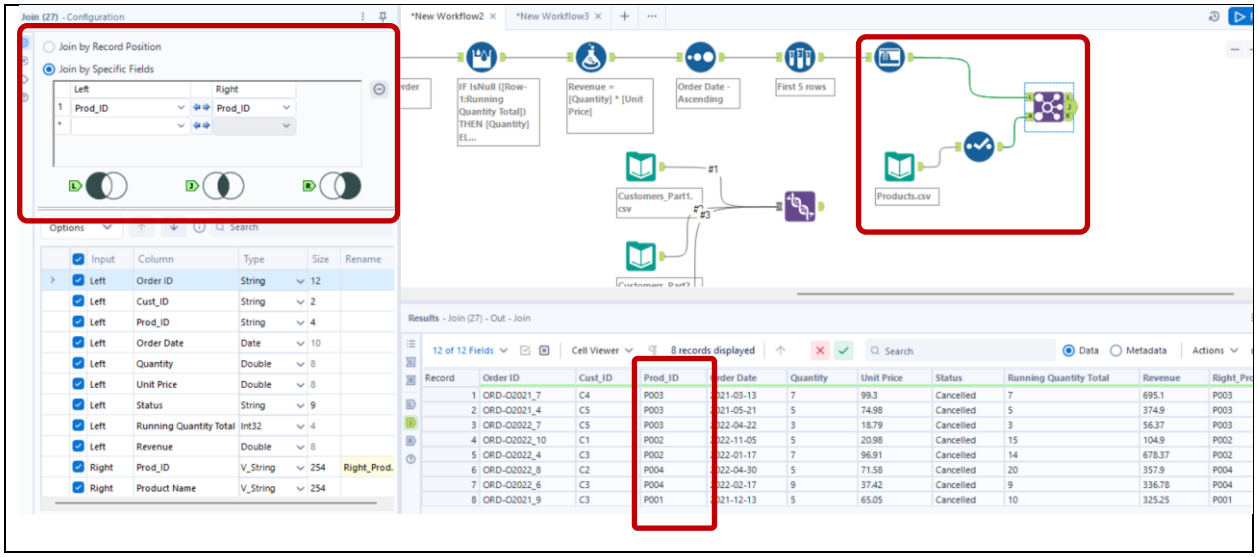
- Using Customers_Part1, Customers_Part2, Customers_Part3.csv dataset.
- Connect outputs of Customers_Part1.csv, Customers_Part2.csv, and Customers_Part3.csv into a Union Tool.
- Use Auto Config by Name setting.



n) Use **Join Tool** to Join Orders with Products based on Product ID (1st Join)

Purpose tool: to combine two inputs based on common fields between 2 tables.

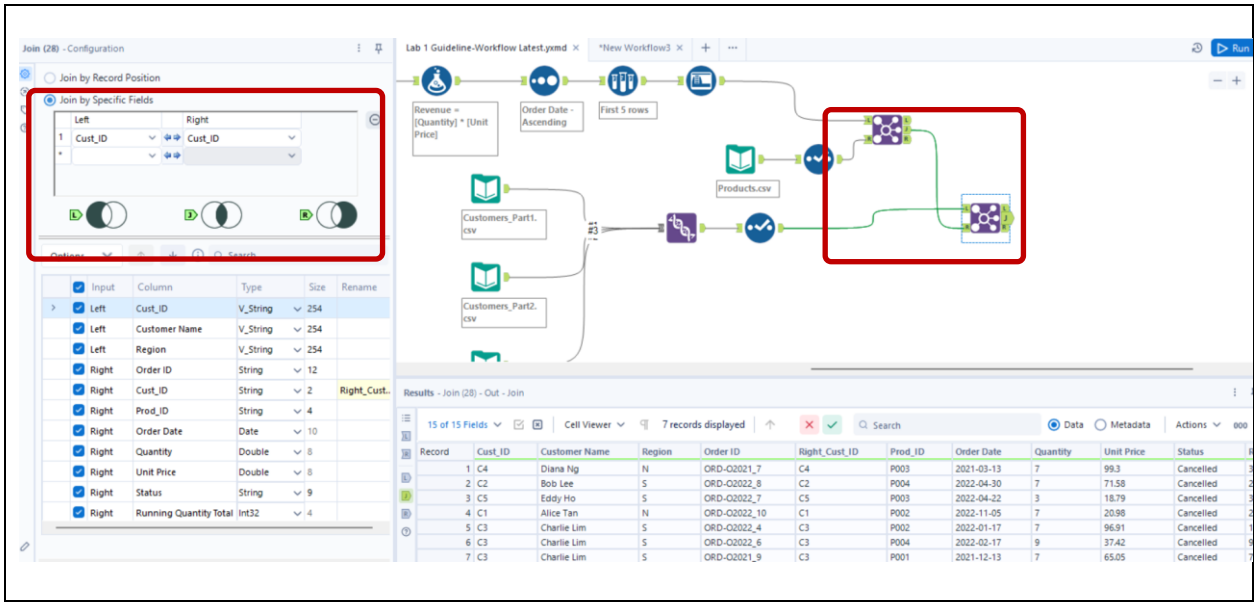
- Join Orders with Products.csv using Product ID
- Left Input: Orders table (already cleaned and prepared).
- Right Input: Products.csv
- Join on: Product ID



o) Use **Join Tool** to Join the result of **1st Join** with **Unioned Customers** based on **Customer ID** (2nd Join).

Purpose tool: to combine two inputs based on common fields between 2 tables.

- Right Input: Output from the first Join (Orders + Products).
- Left Input: Combined Customers dataset from Union.
- Join on: Customer ID



p) Use **Find Replace Tool** to Replace short region codes (N, S) with full names (North, South).

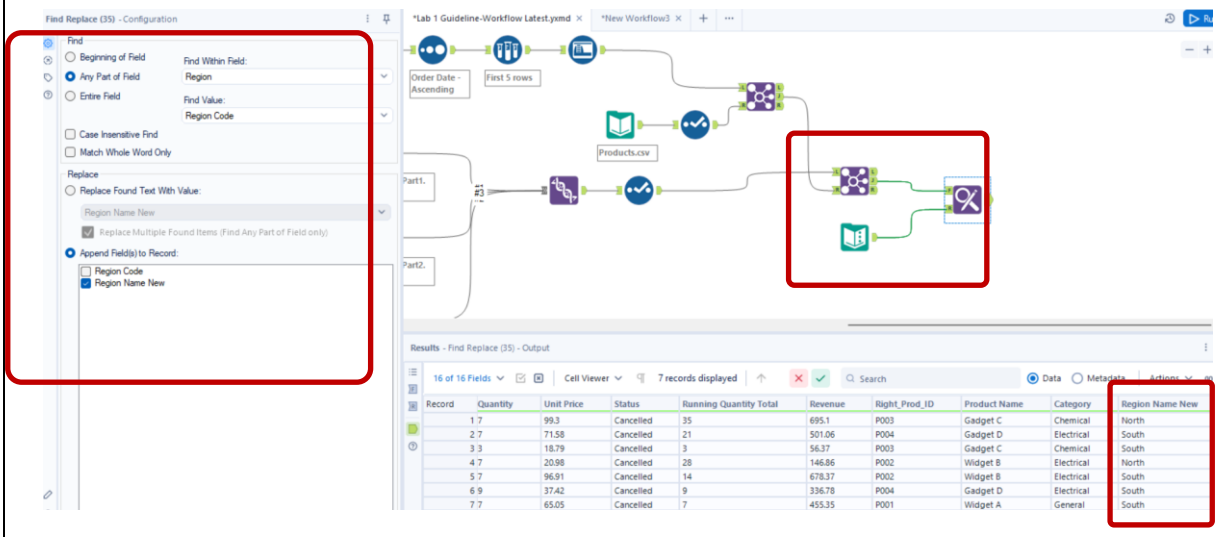
Purpose tool: to update value using a reference list.

- Using Customer.xlsx dataset.
- Create a lookup table:
  - ➢ Use **Text Input Tool** to store value (Region Code: N, S; Region Name News: North, South)
- Input 1: Output from previous join.
- Input 2: Lookup table.
- Match Field: Region
- Output Field: Region Name (Append as new field)

OR

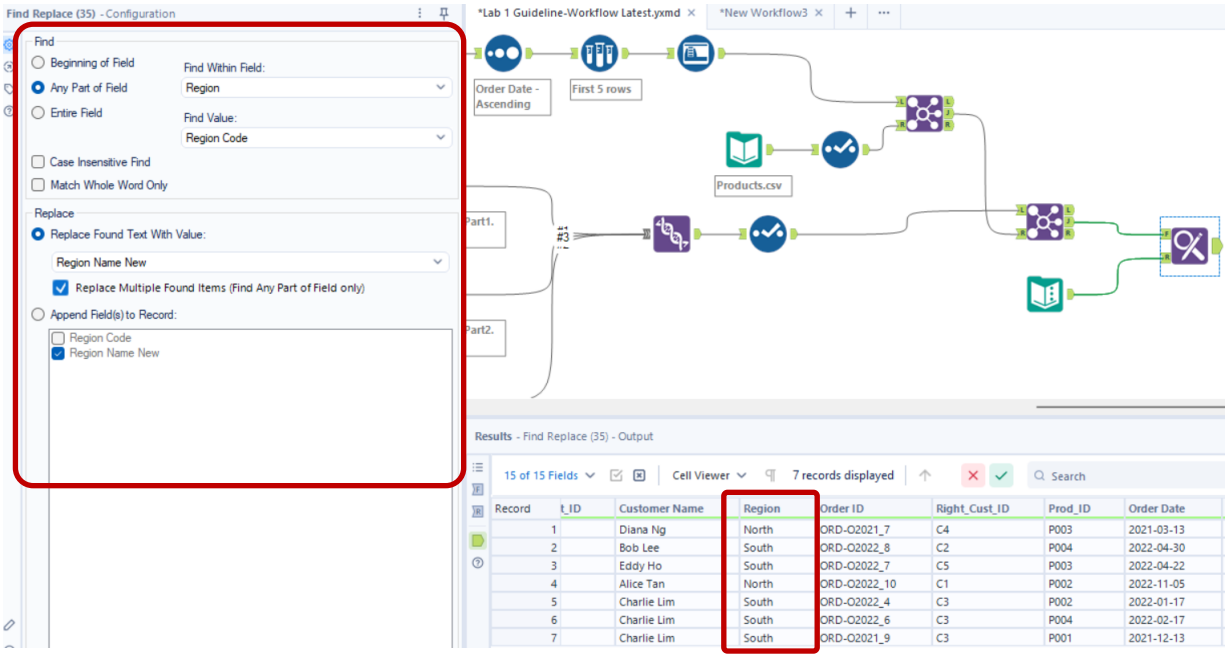
- Using formula:
  IF [Region] = "N" THEN "North"
  ELSEIF [Region] = "S" THEN "South"
  ELSE "Unknown"
  ENDIF

If Append



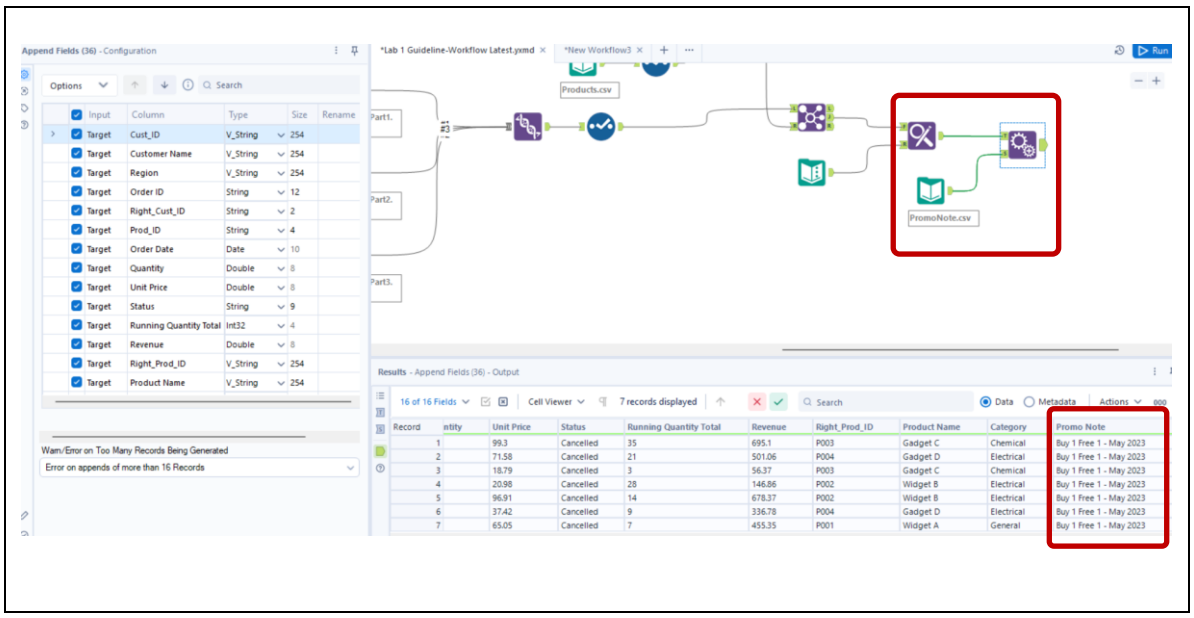
OR

If Replace



q) Use **Append Fields Tool to add a promo message from PromoNote.csv to every record**.
Purpose tool: to update value using a reference list.

- Left Input: Output from Find Replace.
- Right Input: **PromoNote.csv** (Promo Note field: "Buy 1 Free 1 Promo").



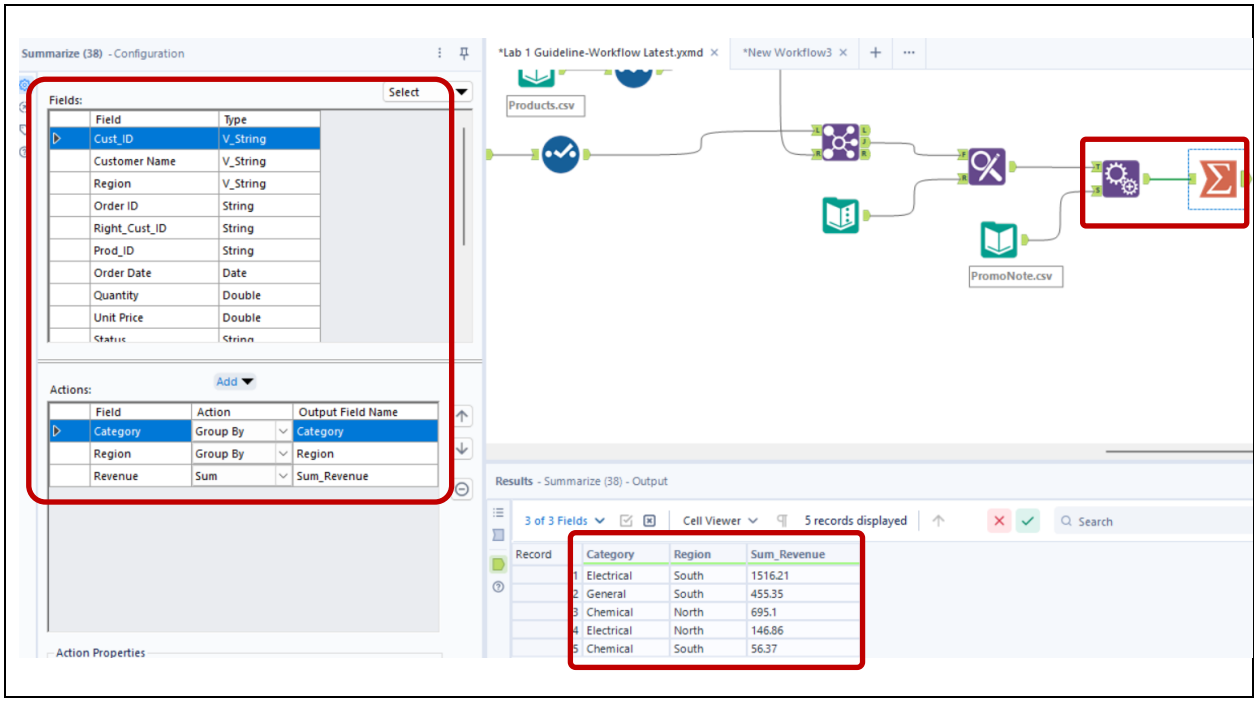
Step 5: Transform Data

Purpose: Aggregate and group the data into meaningful insights.
Tools: Summarize, Transpose, Cross Tab

r) Use **Summarize Tool** to Group orders by Category and Region Name and calculate the Total Revenue.

Purpose tool: to perform various actions (functions and calculations) on our data.

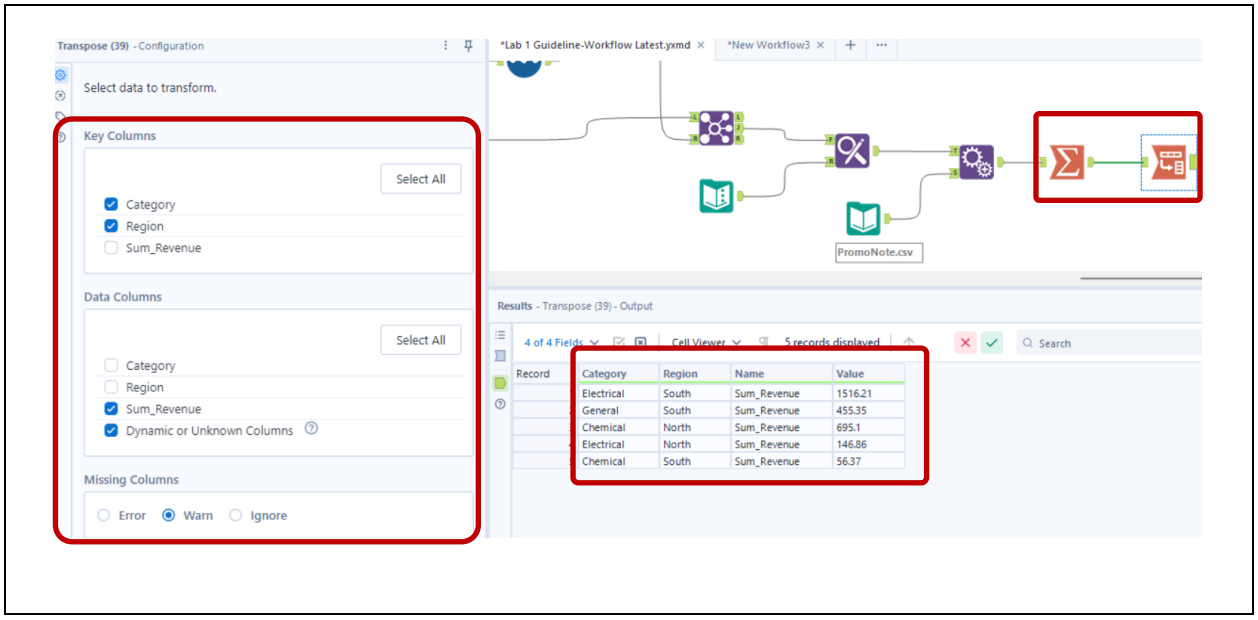
- Drag a Summarize Tool onto the canvas.
- Connect it after your cleaned and joined dataset.
- Configuration inside Summarize:
  - ➢ Group By: Category
  - ➢ Group By: Region
  - ➢ Sum: Revenue.



s) Use **Transpose Tool** to Reshape the summary so that fields become row-wise data.

Purpose tool: to pivot the orientation of the data table. This tool is the reverse application of the Cross Tab Tool.

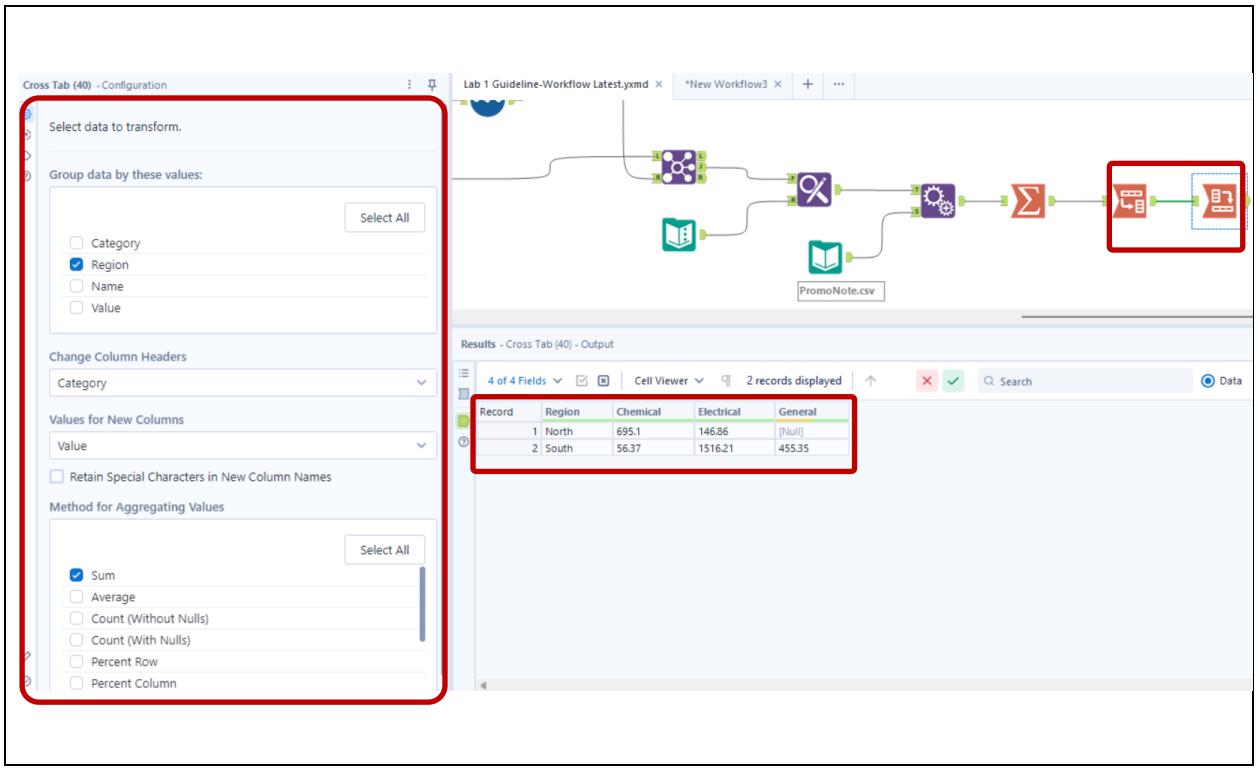
- Drag a Transpose Tool
- Configuration inside Transpose:
  - ➢ Key Fields: Category, Region
  - ➢ Data Field: Total Revenue



t) Use **Cross Tab Tool** to make Pivot data: Make **Region Name** as rows, and **Category** as columns.

Purpose tool: to pivot the orientation of the data in a table by moving vertical data fields onto a horizontal axis and summarizing data where specified.

- Drag a Cross Tab Tool.
- Configuration:
  - Group Data By: Region Name
  - Change Column Headers: Category
  - Values for New Columns: Sum of Total Revenue

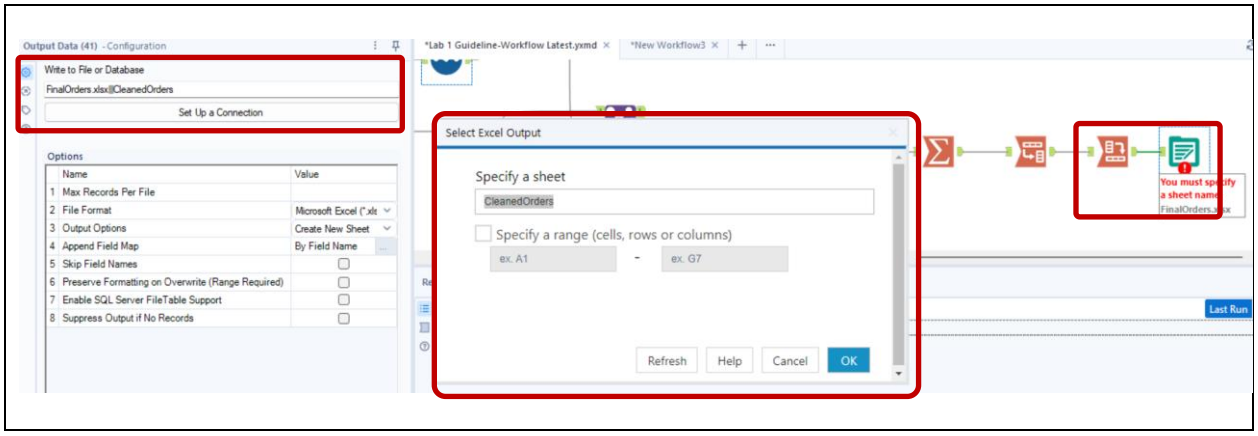


Step 6: Report

Purpose: Present the results clearly and professionally.
Tools: Output Data, Table, Report Header, Render

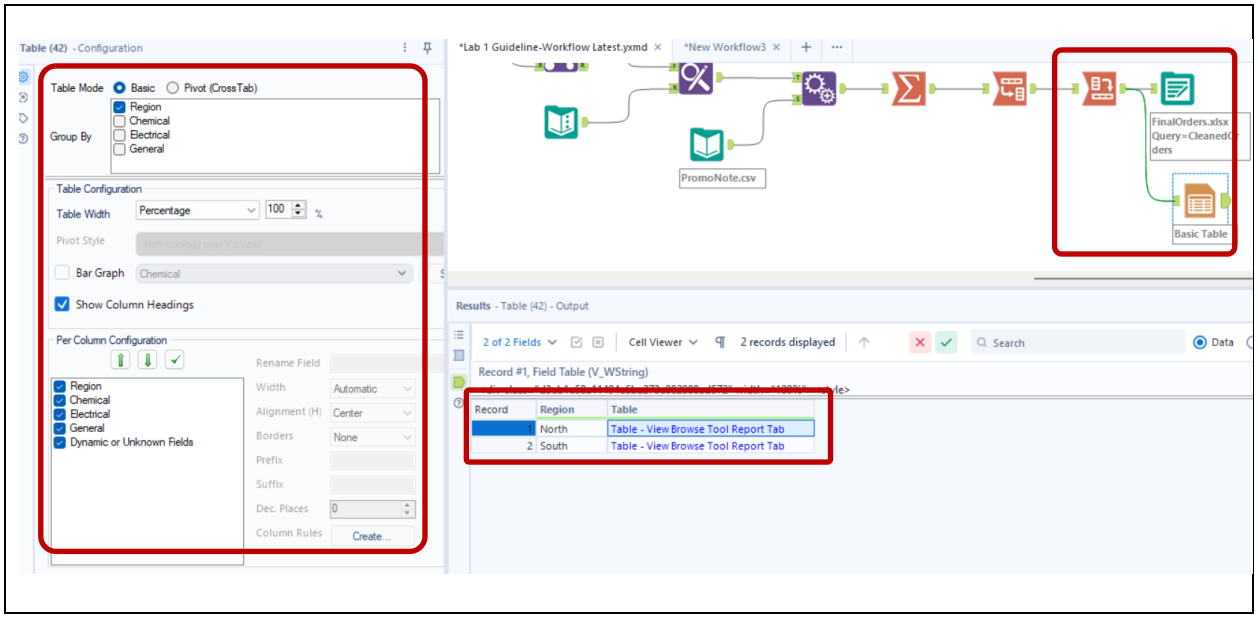
u) Use **Output Data Tool** to Save the final cleaned dataset into an Excel file.
Purpose tool: to write result of a workflow to support the files types or data sources.

- Drag an Output Data Tool.
- Save as:
  - Output File: FinalOrders.xlsx
  - File Format: Microsoft excel (*.xlsx)
  - Sheet name: CleanedOrders OR
  - C:\Users\YourName\Documents**FinalOrders.xlsx|||CleanedOrders**



v) Use **Table Tool** to format your summarized data into a nice clean report table.
Purpose tool: to create a data or pivot table to output in a report via the Render tool.

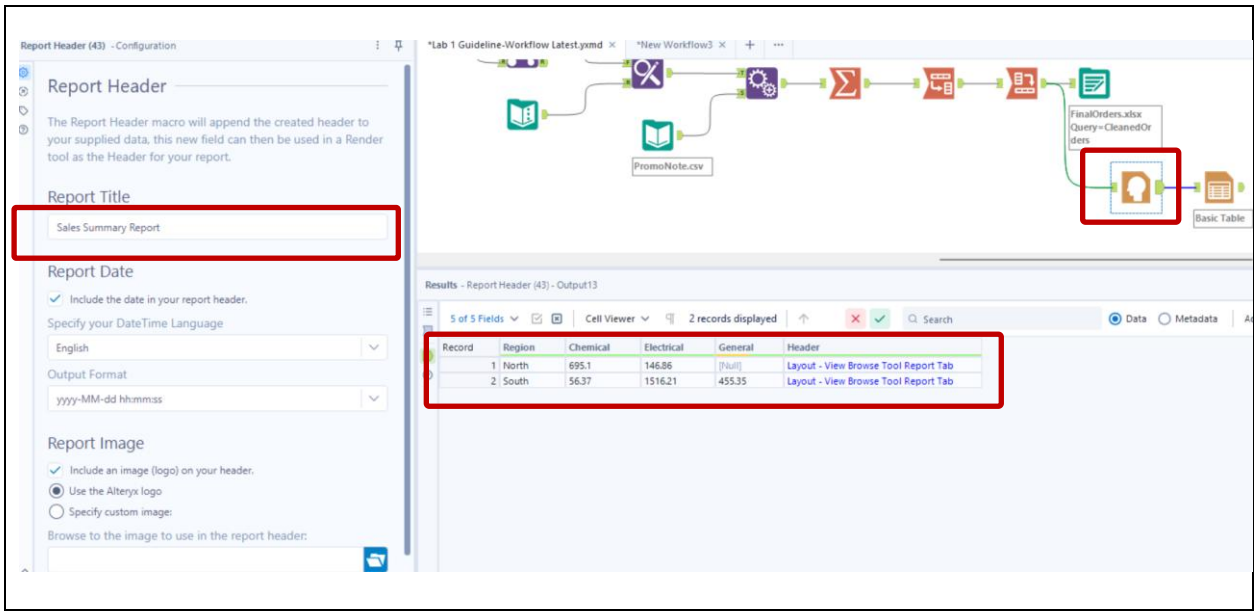
- Drag Table Tool.
- Connect it after the Cross Tab output.
- Customize column headers and style if needed.



w) Use **Report Header Tool** to add a professional title to the report.

Purpose tool: to setup and place a header onto your report.

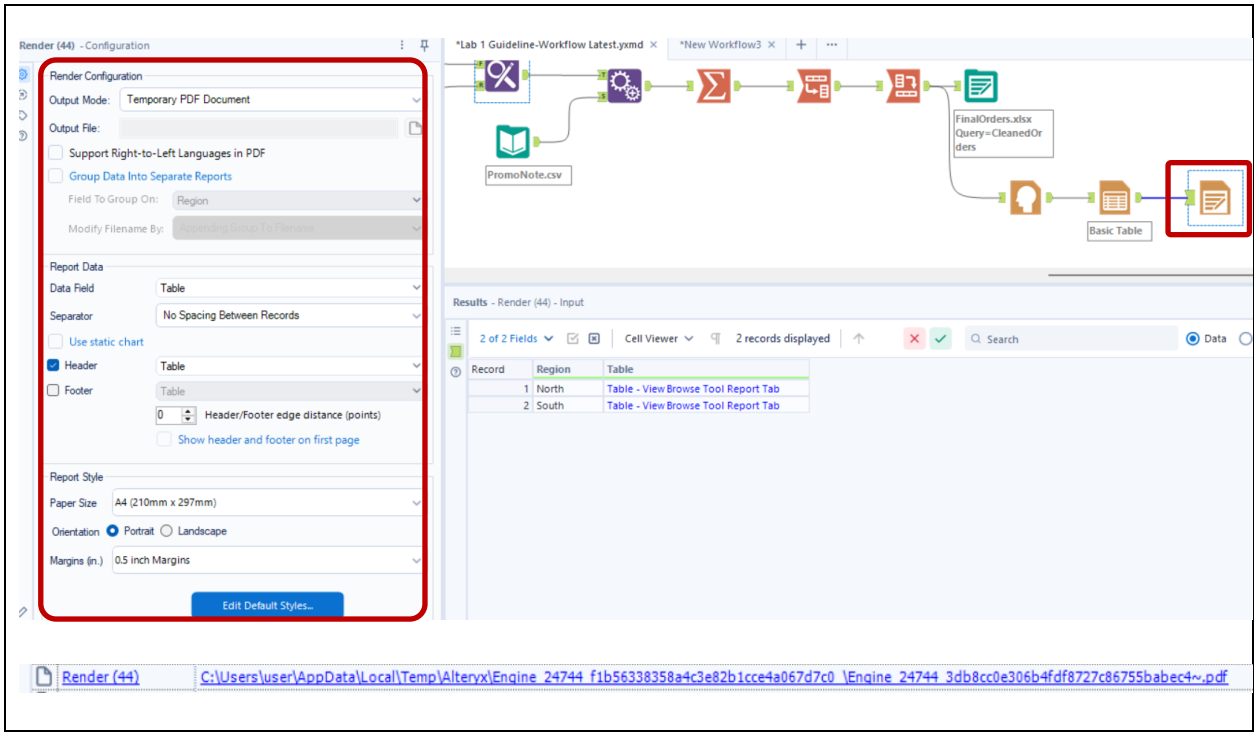
- Drag Report Header Tool before the Table Tool.
- Set title:
  - "Sales Summary Report"

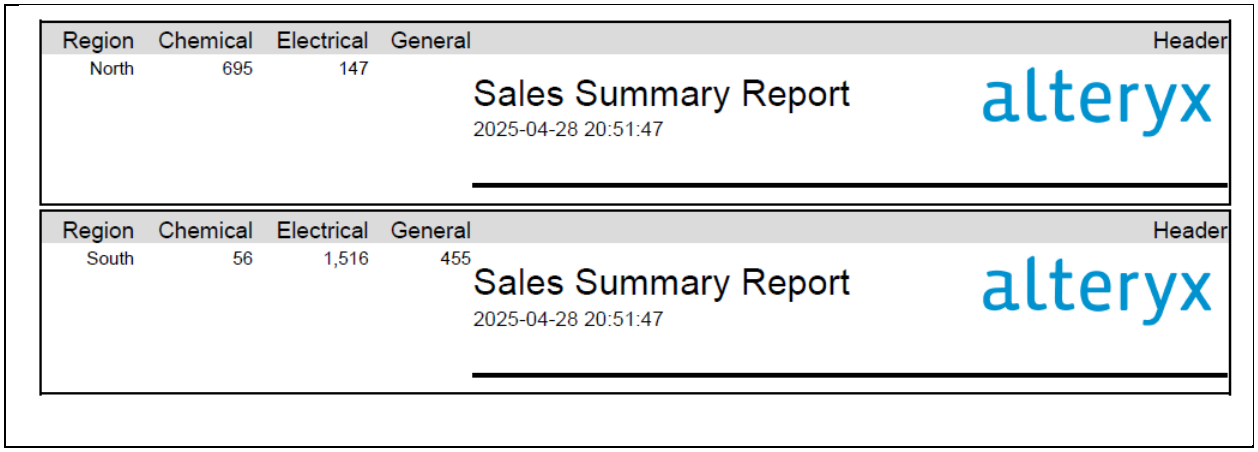


x) Use **Render Tool** to Export the final report to PDF format.

Purpose tool: to transform report snippets into presentation quality reports in a variety of format. Output to Microsoft office format is only supported for Office 2007 and above.

- Drag a Render Tool.
- Connect after the Report Header + Table
- Save as:
  - " SalesSummaryReport.pdf"




Region Chemical Electrical General Header
North 695 147
Sales Summary Report
2025-04-28 20:51:47
alteryx
Region Chemical Electrical General Header
South 56 1,516 455
Sales Summary Report
2025-04-28 20:51:47
alteryx